

- 1 A learning institute wants to create its own content management system (CMS) for its students' self-directed learning needs. Students can enrol in as many courses as they wish. They are also required to complete a project by the end of each course.

Each course is conducted by a course coordinator. Each course coordinator may have more than one courses under his/her management.

As part of the system, a relational database needs to be designed.

- (a) The database needs three tables to store the data for the students' enrolment for the courses: `STUDENT`, `ENROLMENT` and `COURSE`.

Draw an Entity-Relationship (E-R) diagram showing the three tables and the relationships between them. [2]

- (b) The database needs four tables to store the data for the courses' registration for the course coordinator: `COORDINATOR`, `PROJECT`, `ENROLMENT` and `COURSE`.

Draw an Entity-Relationship (E-R) diagram showing the four tables and the relationships between them. [3]

- (c) Draw the overall Entity-Relationship (E-R) diagram showing the above tables and the relationships between them. [1]

- (d) Each student is assigned to a unique student ID by the system. To enrol in the course, each student must provide his/her name, contact number and email address.

Each course must have an appropriate name and is assigned to a unique course code. At the end of each course, each student will be given a grade for his/her project. Each project is also given a unique project ID to identify the project to the student.

The course coordinator is also assigned to a unique coordinator ID by the system. The student can contact the course coordinator by email.

The `ENROLMENT` table description has been done for you. Based on the above information, write the table descriptions for the other four tables, each of which are in third normal form (3NF). [4]

```
ENROLMENT (StudentID, CourseCode, DateEnrolled, FinalGrade)
```

The primary key is indicated by underlining one or more attributes. The foreign key is indicated by drawing a dashed line below one or more attributes.

- (e) Singapore's Personal Data Protection Act (PDPA) protects data of all students stored in the database. Describe one way in which PDPA achieves this purpose. [2]

2 Cyberattacks have become more prevalent due to the advent of technology.

(a) With the aid of a diagram, explain how a denial of service (DOS) attack can disrupt a server's operations. [2]

(b) State another malware that can also disrupt a server's operations and explain how it does so. [2]

(c) Network operational devices such as firewalls, intrusion detection system (IDS), and intrusion prevention system (IPS) can be used to counter attacks against a network.

Explain how a firewall is deployed to block unwanted traffic from accessing a particular network. [2]

(d) Often, intrusion detection system (IDS) and intrusion prevention system (IPS) collectively known as IDS systems are used in tandem with firewall for better detection of network attacks. Explain how IDS and IPS help in providing an additional layer of protection for a network. [2]

3 For effective communication to take place, protocols need to be established. Likewise, in a network, protocols are needed.

(a) Explain what is meant by a communication protocol and how it is used to handle the complexity in transmitting data over the Internet. [2]

(b) The File Transfer Protocol (FTP) is used for file transfer from one computer to another while the Simple Mail Transfer Protocol (SMTP) is used to send and receive email.

State the name of the protocol that is used to access the different web pages on the Internet. Briefly describe how this protocol works. [3]

(c) There are two types of computer network architectures: Client-server architecture and peer-to-peer architecture.

Explain why a client-server architecture is implemented in the case of a web application. [2]

(d) Describe how Domain Name System (DNS) allows access to a website using a Uniform Resource Locator (URL). [3]

- 4 The pseudocode below shows a recursive linear search algorithm that returns the index positions of all occurrences of target in a list.

```

01 FUNCTION linear_search(A, target, i ← 0): RETURN LIST
02   position ← []
03
04   IF i < length of A
05     IF A[i] = target
06       THEN
07         Append i to position
08       ENDF
09     position ← position + linear_search(A, target, i+1)
10   ENDF
11   RETURN position
12 ENDFUNCTION

```

(a) Discuss whether iterative linear search is better than recursive linear search. [2]

(b) The following is the implementation of quick sort in Python:

```

01 def quicksort(array):
02     if array == []:
03         return []
04     else:
05         pivot = array[0]
06         less = []
07         great = []
08         for i in array[1:]:
09             if i < pivot:
10                 less.append(i)
11             else:
12                 great.append(i)
13         less = quicksort(less)
14         great = quicksort(great)
15         return less + [pivot] + great

```

(i) Given the array [9, 12, 9, 2, 17, 1, 6], trace how the items are sorted using quicksort. [4]

(ii) Evaluate and justify the circumstances when quicksort should be used and shouldn't be used as the sorting algorithm. [2]

5 A college has two types of students: full-time and part-time.

All students have their name and date of birth recorded.

A full-time student has their address and telephone number recorded.

A part-time student attends one or more courses. A fee is charged for each course. The number of courses a part-time student attends is recorded, along with the total fee and whether or not the fee has been paid.

The college need a program to process data about its students. The program will use an object-oriented programming language.

(a) Draw a class diagram that shows the following for the situation described above:

- the superclass
- any subclass
- inheritance
- properties
- appropriate methods

[7]

(b) State the purpose of encapsulation.

[2]

A part-time student can be either a local student or a foreign student. A foreign student will need to pay an additional administration fee for each enrolled course.

(c) Suggest a change to the class diagram to enable the new calculation of total fee used for foreign part-time student.

[1]

6 The university wants to assess troubleshooting issues with a printer. It wants to use a decision table to do this.

The troubleshooting actions are:

- check the connection from computer to printer if the error light is flashing and the document has not been printed
- check the ink status if the quality is poor
- check whether there is a paper jam if the error light is flashing and the document has not been printed
- check the paper size selected if the paper size is incorrect.

(a) Create a decision table to show these conditions and actions.

[4]

Conditions	Document printed but quality is poor
	Error light flashing on printer
	Document printed but paper size is incorrect

(b) Simplify your decision table by removing redundancies from the decision table.

[2]

- 7 A binary search tree is implemented using an array, `Names`. The names are stored in the elements of the array in alphabetical order. Each element of the array `Names` comprises three parts: a left pointer, the name and a right pointer.

<code>l_ptr</code>	<code>Name</code>	<code>r_ptr</code>
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The pointers contain the array index of a node to either the left or right of the current node. -1 indicates there are no further nodes in a particular direction.

The root of the binary search tree is stored in an integer variable, `root`.

Another integer variable, `free_pointer`, stores the position of the next free node. All free nodes are stored in a linked list.

The contents of the array `Names` are shown:

<code>free_pointer</code>	<code>Index</code>	<code>l_ptr</code>	<code>Name</code>	<code>r_ptr</code>
5	0	-1	Celine	-1
	1	-1	Elli	-1
	2	3	Ben	0
	3	-1	Ali	-1
	4	2	Dodi	6
	5	-1	None	8
	6	1	Farai	7
	7	-1	George	-1
	8	5	None	-1

- (a) Draw the binary search tree represented by the data in the array `Names`. [2]
- (b) A new name, **Faith**, is to be inserted into the binary search tree. Show the changes to the array `Names` after this insertion. [3]
- (c) State the index of `free_pointer` after the insertion in part (b). [1]
- (d) A programmer needs an algorithm to output the result of an in-order traversal of `Names`. To design this, the programmer writes a recursive procedure in pseudocode.

```

01 PROCEDURE TraverseTree(BYVALUE Root: INTEGER)
02   IF Tree[Root].l_ptr .....A.....
03   THEN
04     TraverseTree(.....B.....)
05   ENDIF
06   OUTPUT .....C..... .Name
07   IF .....D..... <> -1
08   THEN
09     TraverseTree(.....E.....)
10   ENDIF
11 ENDPROCEDURE

```

Complete the pseudocode for A, B, C, D and E in the algorithm.

[5]